



Arab American University

Software Engineering – Fall 2025

EduFocus System

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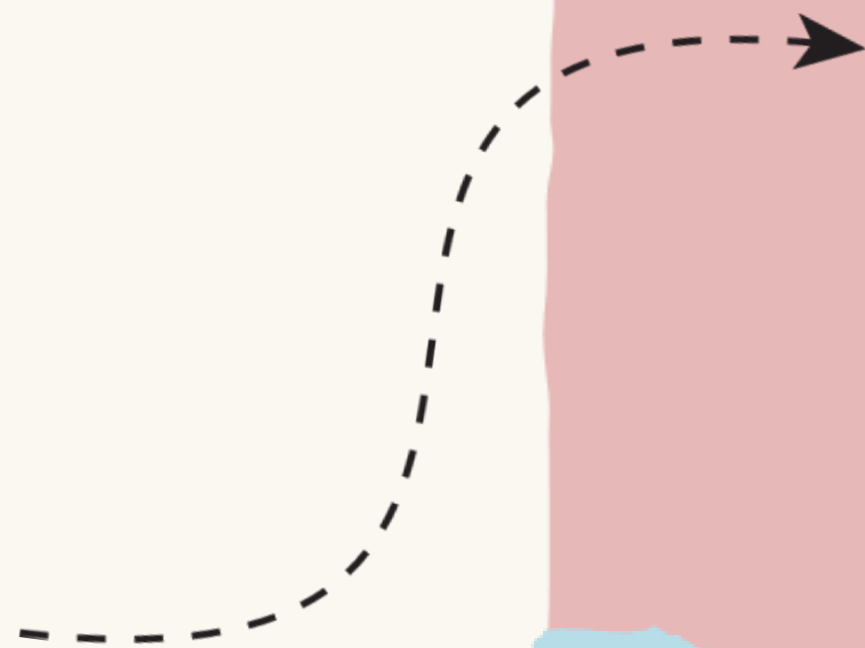
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Introduction

EduFocus is a web-based system designed to improve student focus and participation during online and blended university lectures. The system addresses the problem of student distraction in virtual learning environments by providing real-time attention feedback in a simple and practical way, while respecting privacy and ethical guidelines.



Problem Statement

Many students face difficulties staying attentive in virtual learning environments due to distractions, and this system aims to address that issue in a simple and practical way.

Stakeholders

- Students
- Instructors
- University
- Ministry of Education
- System Administrators

Impact

gap in the quality of education due to lower student performance, and Reduced learning effectiveness.

Solution

EduFocus provides an automated solution that monitors focus in real time and presents the results in a clear and simple way. The system is intended to be secure, easy to use, and accessible to different types of users while respecting privacy and ethical guidelines.

Goals



The system focuses on increase students' awareness of their attention levels during lectures.

Reduce distraction in online and blended learning environments.

Objectives



The system helps students know how focused they are during lectures in a quick and non-disruptive way. At the same time, it gives teachers overall reports on student engagement without showing any individual student's data. The system is designed to protect privacy, be used ethically, not disturb learning, and work easily with existing university tools.

Main Contributions

Our System **Contribution in Education** by Reduces the engagement gap in remote learning via providing instant feedback to both students and teachers, This is therefore considered a **Societal Benefit**, because it Supports the national digital transformation in education and helps students with the issue of attention and focus.



similar systems



During our research into similar systems, we discovered:
"Intel & Classroom Technologies" a collaboration in China and the United States that uses artificial intelligence to monitor students' emotions and focus via camera.

How does the "EduFocus" project differ from them?

Privacy-Centric: In global systems, video is often stored or analyzed on servers, while "EduFocus" processes data locally and does not record any video. This addresses a common concern with this technology: the fear of surveillance.

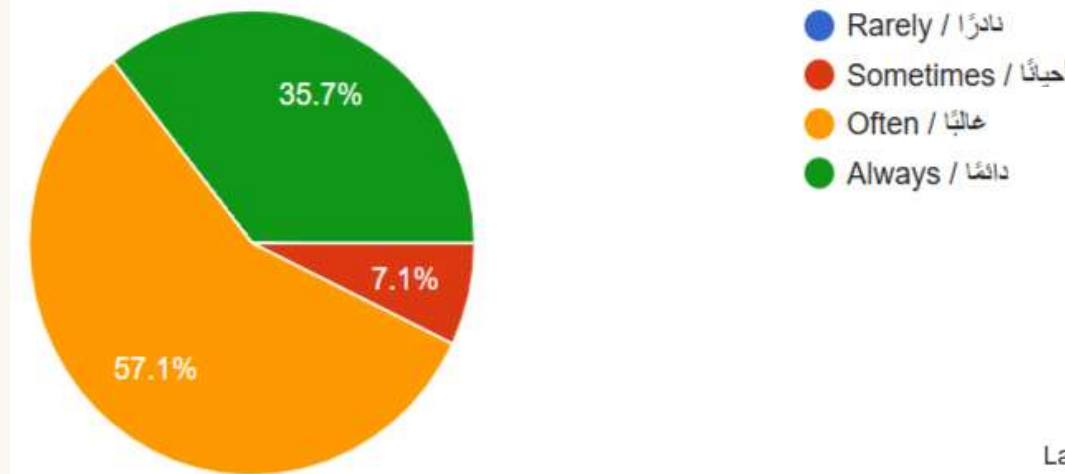
Standalone System: Our project is designed to be a standalone system that can be used with any educational platform, not limited to a single program.

Gentle Nudges: The focus is on providing students with gentle reminders to help them self-monitor, rather than using a monitoring (punishment) approach.

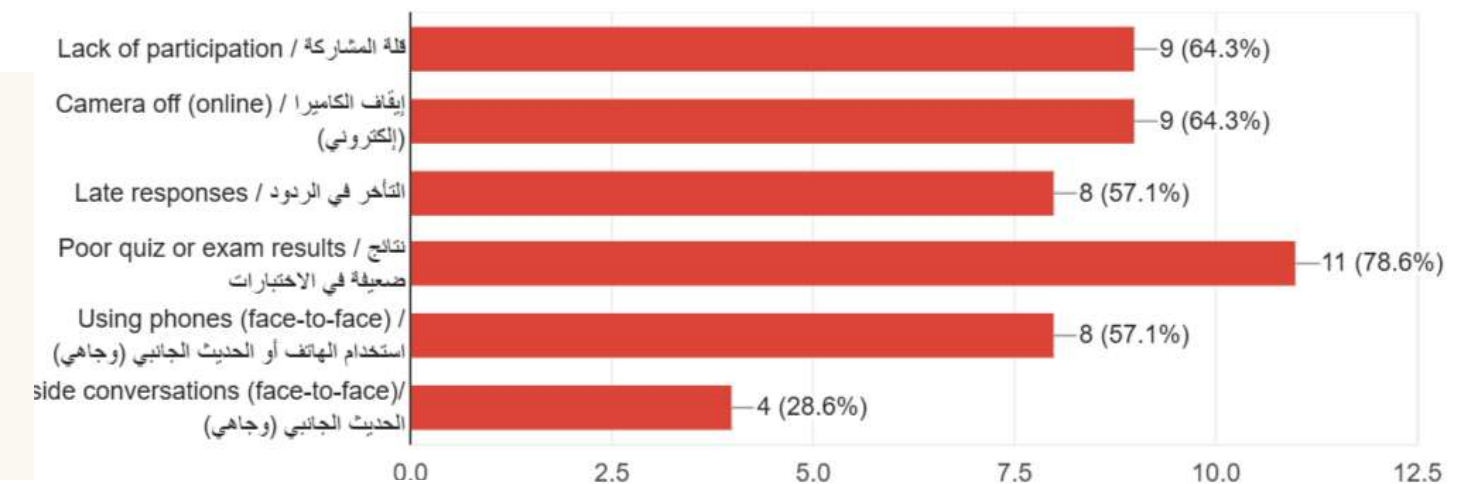
Methodology

We adopted a structured engineering methodology to ensure the system's accuracy. We began with **elicitation**, where we conducted a **survey** and distributed it to students and instructors.

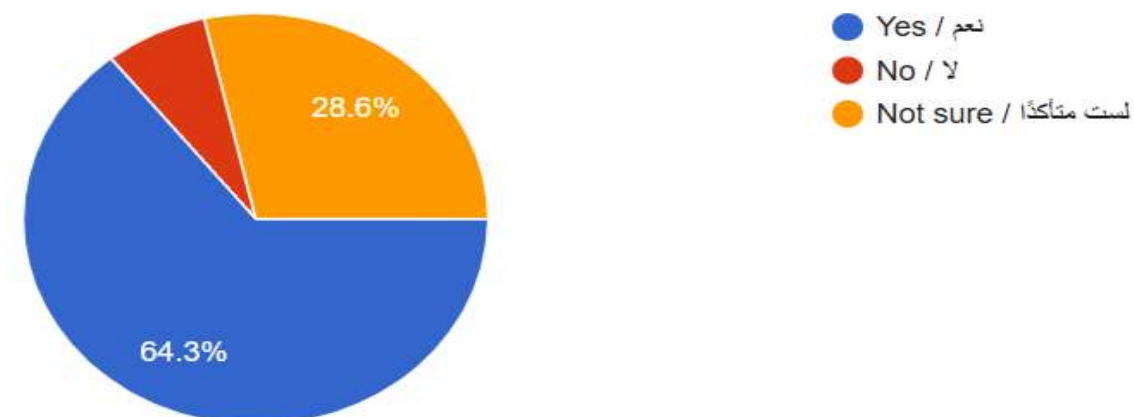
?Do you notice students losing focus during lectures (online or face-to-face)
هل تلاحظ فقدان الطلاب للتركيز أثناء المحاضرات (الإلكترونية أو الواجهية)؟
(اختيار واحد / Multiple choice)



?What indicators show that students are distracted
ما المؤشرات التي تدل على تشتت الطلاب؟
(اختيار متعدد / Checkboxes)



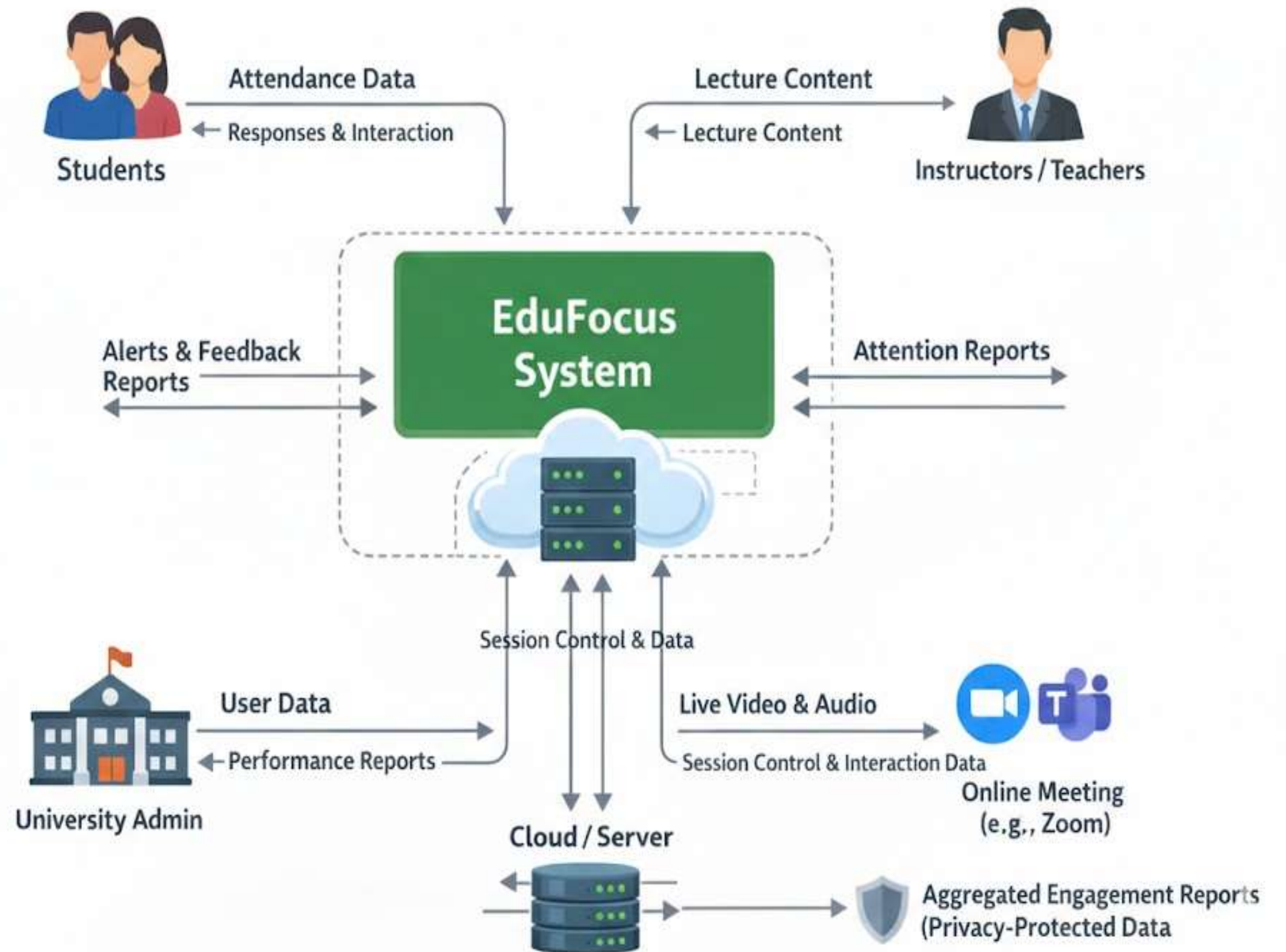
?Do you think monitoring student attention improves learning outcomes
هل تعتقد أن مراقبة انتباه الطلاب تحسن نتائج التعلم؟
(اختيار واحد / Multiple choice)



Methodology

Next, we moved to the analysis phase, where we created a **context diagram** to precisely define the system's boundaries.

Then we defined the requirements (Specification) in the SRS document based on international standards, and finally the validation stage.



System Operation & Attention Analysis Flow

01

User Consent

User approves attention tracking before the lecture starts to ensure privacy and ethical use.

02

Data Capture (Frames)

The system captures temporary camera frames during the lecture for real-time analysis only (no storage).

03

Local AI Processing

Attention analysis is performed locally on the user's device using **TensorFlow.js**.

04

Focus Score Calculation

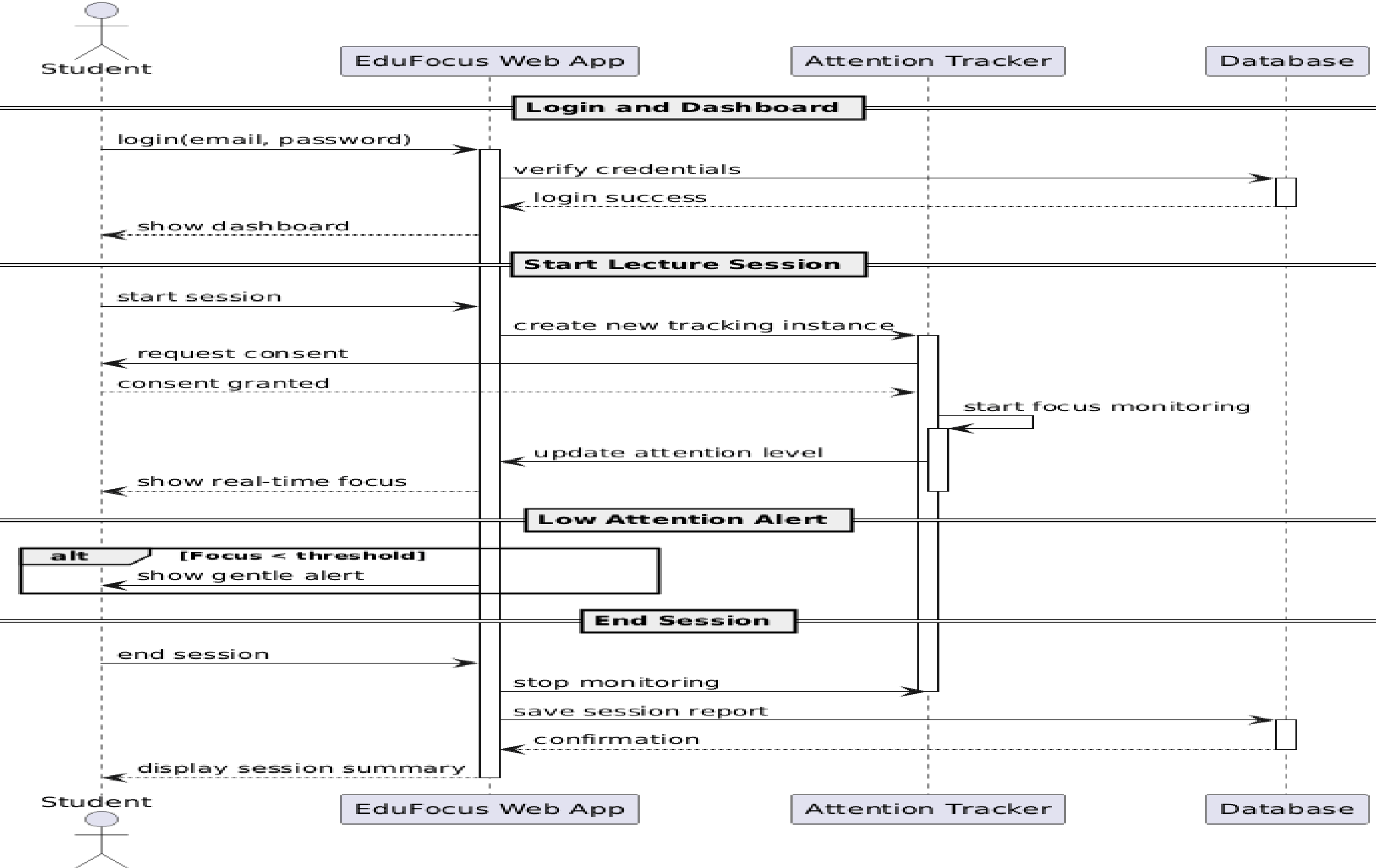
Visual indicators are analyzed to compute a focus score (**High / Medium / Low**).

05

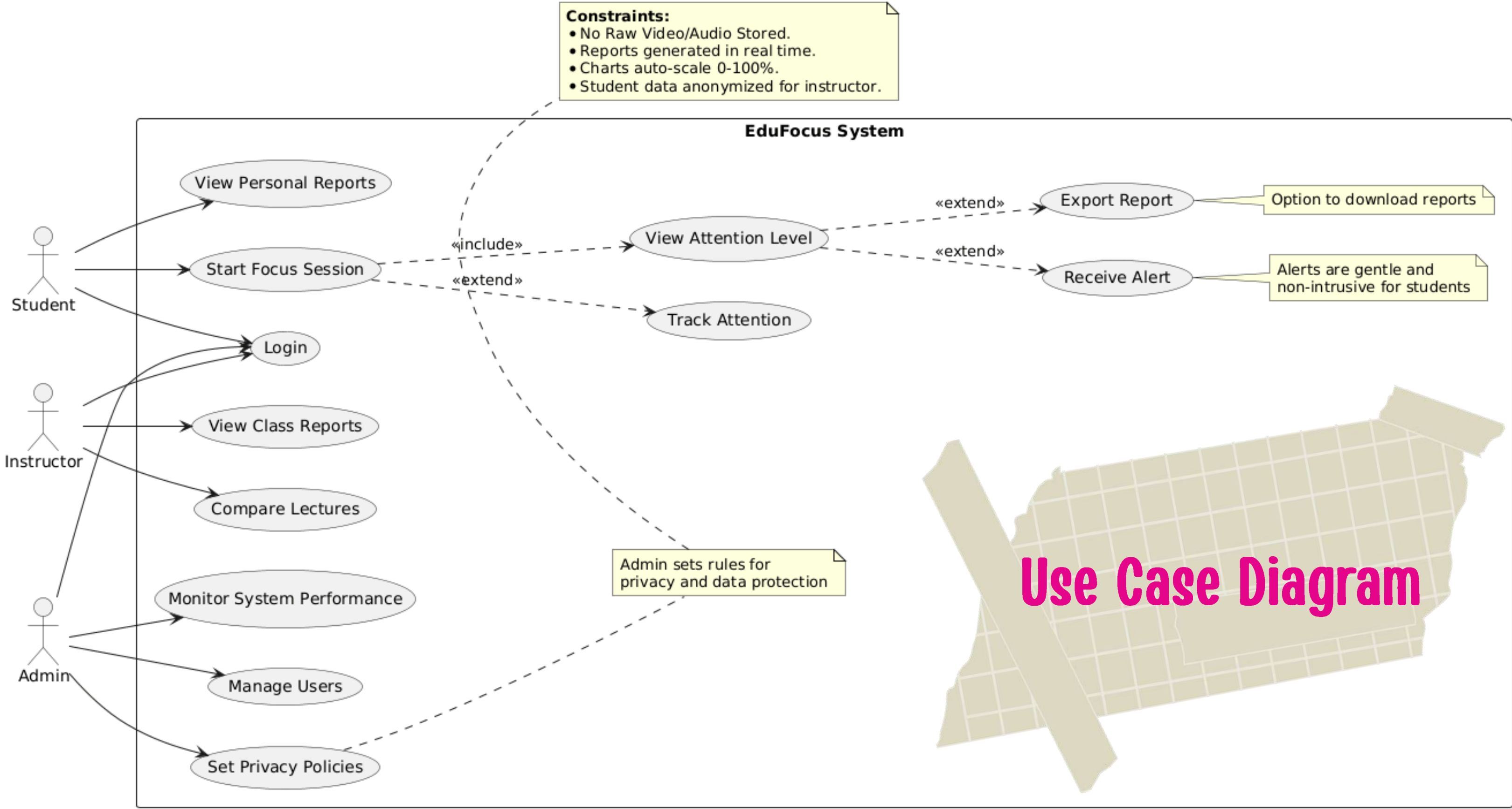
Alerts & Reports

The system sends gentle real-time alerts and generates summarized, anonymized reports after the lecture.

Sequence Diagram



System Modeling



User Authentication and Role Management:

- Allows users to log in and access system features based on their role.

Real-Time Attention Data Collection and Analysis

- Collects and analyzes data of attention during lectures in real time.

Attention Level Display and Alerting

- Displays the current attention level to students and sends gentle alerts when focus drops.

Student Attention Reports and History

- Reports are generated after the lecture to help the student review and improve their focus.

Instructor Dashboard and Class Engagement Reports

- Reports for lecturers showing student interaction and focus during the lecture (feedback)

System Administration and Privacy Control

- Enables administrators to manage users, settings, and privacy policies

Functional Requirements

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graph TD; FR[Functional Requirements] -.-> UAR[User Authentication and Role Management]; FR -.-> RTAD[Real-Time Attention Data Collection and Analysis]; FR -.-> SARH[Student Attention Reports and History]; FR -.-> SAPC[System Administration and Privacy Control]; FR -.-> ID[Instructor Dashboard and Class Engagement Reports]; FR -.-> ALDA[Attention Level Display and Alerting];
```

Non-Functional Requirements

Simple and intuitive interface across different devices.

Usability

ensures successful detection of attention and reliable generation of reports

Accuracy

processing in real time with quick answer and delay in minimum.

Performance

Lecture's performance should be reliable and available(parameters)

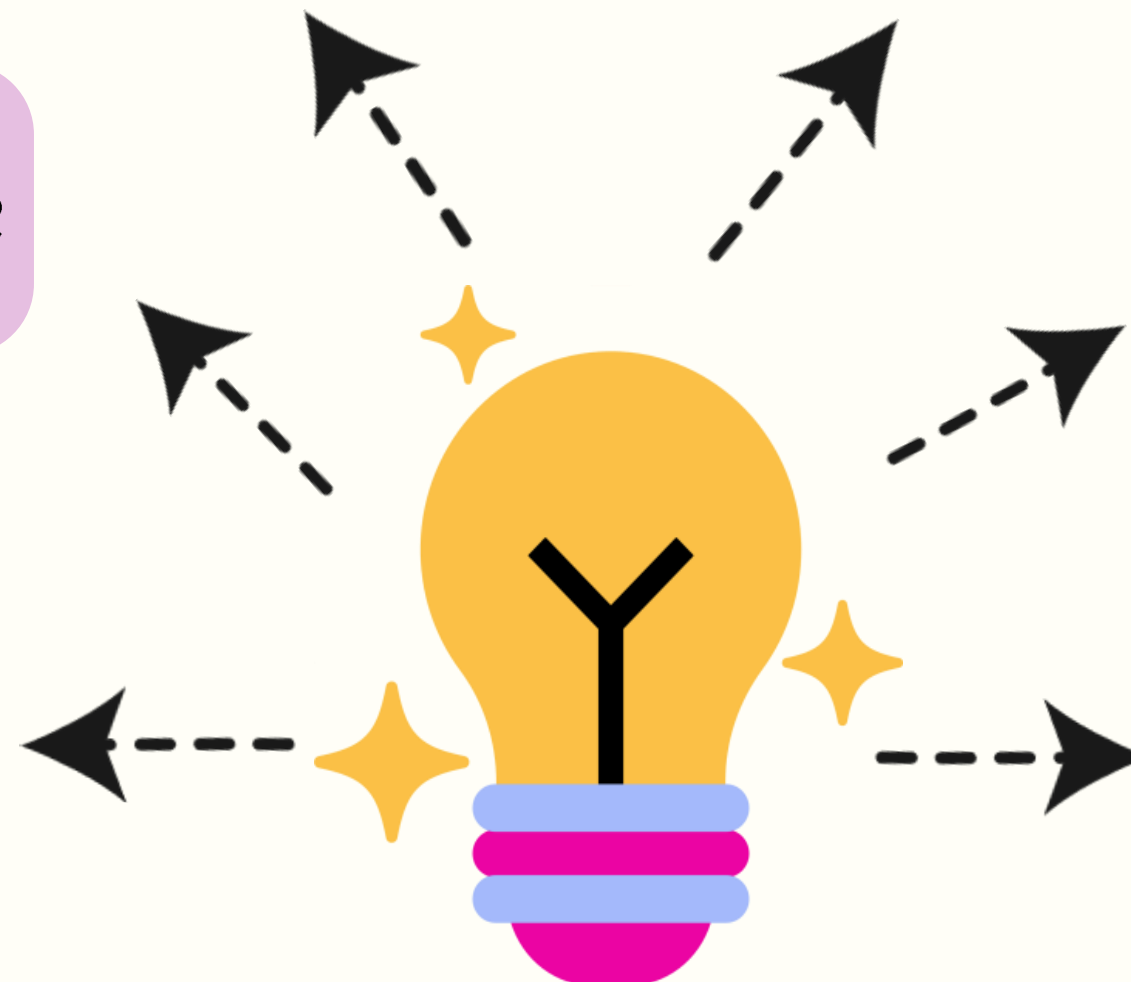
Reliability & Availability

privacy is our utmost priority. We don't save your video or audio.

Security & Privacy

Operates with laptops, mobile phones and wearables.

Compatibility



Conclusion and Future Work



We can now say that we have succeeded in designing a comprehensive system that monitors student attention using artificial intelligence while ensuring complete privacy. In the future, we will try to further develop the system by incorporating AI tools and integrating it into more tangible devices.

In the future, we will try to further develop the system by:

Improving AI capabilities: Enhancing the accuracy of the existing attention analysis model and adding adaptive machine learning techniques.

Wearable Integration: Expanding support for more types of devices, i.e., integrating it into more tangible devices.



THANK
YOU!

